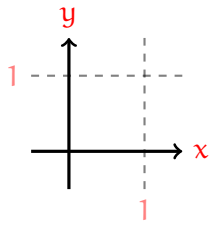
Example 1. Find the following by converting to spherical coordinates

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{1+\sqrt{1-x^2-y^2}} 1 \, dz dy dx$$

Solution. The outer bounds determine the xy–shadow.



This indicates that the constant range of θ is:

Let us rewrite the inner bounds in spherical.

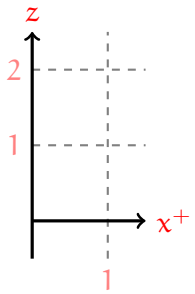
$z = 1 + \sqrt{1 - x^2 - y^2}$

$z = \sqrt{x^2 + y^2}$

Neither of these equations involves:

So the region is obtained by rotating its x^+z–cross section

Let's sketch its x^+z cross–section.



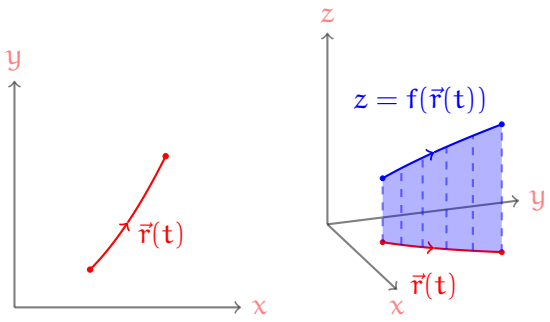
So for fixed θ from $0 \leqslant \theta \leqslant \pi/2$, the range of ϕ is:

And for fixed (θ, ϕ) from the $\theta\phi$–shadow, the range of ρ is:

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{1+\sqrt{1-x^2-y^2}} 1 \, dz dy dx =$$

A. **Scalar Line Integrals.** We integrate on curves. One possible ingredient set is

We sketch the "fence" with base along the path and height determined by f.



To find the fence area: first we find a small bit of **arclength** ds along our path.

$ds =$

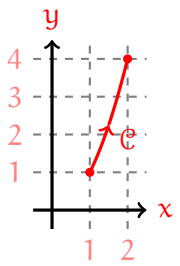
The area of an infinitesimally thin vertical panel of our fence is

[area of panel with base ds] $=$

The **scalar line integral** of f along the path $\vec{r}(t)$ with $a \leqslant t \leqslant b$ is

[area of fence] $=$

Example 2. Let $\mathcal{C}$ be the segment of the parabola $y = x^2$ from $x = 1$ to $x = 2$.



Find $\int_{\mathcal{C}} 8x \, ds$.

Solution. First we parametrize the curve.

Because $y = x^2$ defines y in terms of x, a reasonable parameter is x.

$\vec{r}(x) =$

Next we calculate the arclength parameter.

$ds =$

Lastly we write the integral in terms of our parameter.

$\int_{\mathcal{C}} 8x \, ds =$

Example 3. Find the area of the fence above circle $x^2 + y^2 = 4$ in the xy–plane, and below the elliptic paraboloid $z = x^2 + 4y^2$.

Solution. Fence area is a scalar line integral. So first we parametrize.

The curve is the circle $x^2 + y^2 = 4$. So a good parameter is polar θ.

$\vec{r}(\theta) =$

Next we calculate the arclength parameter using $ds = \|\vec{r}'(\theta)\|\, d\theta$.

$ds =$

Lastly we write the integral in terms of our parameter: $x = 2\cos\theta$, $y = 2\sin\theta$.

[fence area] $=$

We use power–reduction formulas: $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$ and $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$.